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Machine Learning Project

Course Project — 2026

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1. Introduction

1.1 Background

Customer churn is one of the biggest problems for telecommunications companies today. Churn is when people stop using a company's service and go to a competitor. In such a competitive telecommunications market, acquiring new customers costs companies 5-7 times more than retaining their current customers. Therefore, predicting customers likely to churn can help companies implement corrective retention strategies before they are lost.

1.2 Problem Definition

This project will develop a machine learning prediction model to determine whether or not a telecommunications customer will churn, based on their demographic information, account information, and service usage patterns.

1.3 Project Objectives

- Determine the important drivers of customer churn
- Develop and compare several different machine learning models
- Measure the performance of the models using appropriate performance metrics
- Develop actionable business intelligence to decrease churn.

1.4 Dataset Information

The dataset used for this analysis is the Telco Customer Churn dataset from Kaggle, which is publicly available. It consists of 7,043 customer records and contains 21 features.

Dataset Summary:

Property	Value
Number of records	7,043
Number of features	21
Target variable	Churn (Yes/No)
Churn rate	26.54%
Data source	Kaggle

Features include:

- **Demographic:** gender, SeniorCitizen, Partner, Dependents
- **Account information:** tenure, Contract, PaperlessBilling, PaymentMethod, MonthlyCharges, TotalCharges
- **Services used:** PhoneService, MultipleLines, InternetService, OnlineSecurity, OnlineBackup, DeviceProtection, TechSupport, StreamingTV, StreamingMovies

2. Methodology

2.1 Machine Learning Workflow

The project utilized an entire Machine Learning process which consisted of the following elements.

Data Gathering → Data Cleaning → EDA → Pre-processing → Train Test Split → Model Training → Reviewing/Insight

2.2 Data Cleaning

The following were completed:

1. TotalCharges converted from String to Numeric (Errors = Coerce).
2. TotalCharges missing data filled in with the average.
3. Removed customer ID (not useful in making prediction).
4. Encoded target (Churn = 1 (Yes) Churn = 0 (No)).

Result: No missing values remain after cleaning.

2.3 Exploratory Data Analysis (EDA)

- Visualizations created to evaluate:
- Distribution of customers that churned by segment.
- Churn and the type of contract.
- Churn by tenure.
- Churned vs. loyal customer characteristics by their monthly charges.
- The impact of Paperless billing on churn.

2.4 Data Pre Processing

Feature Separation:

- Numerical Features: tenure, MonthlyCharges, TotalCharges
- Categorical Features: Remaining Features not including the Target Variable.

Preprocessing Pipeline:

```
# Numerical pipeline
- SimpleImputer (strategy='median')
- StandardScaler


# Categorical pipeline
- SimpleImputer (strategy='most_frequent')
- OneHotEncoder (drop='first')
```

Train-Test Split:

- 80% Training, 20% Testing
- Stratified split to maintain churn distribution

Split	Size	Percentage
Training	5,634	80%
Testing	1,409	20%
Split	Size	Percentage

2.5 Models Selected

Two models were implemented and compared:

Model	Description	Rationale
Logistic Regression	Linear classification model	Simple, interpretable, good baseline
Random Forest	Ensemble of decision trees	Handles non-linear relationships, provides feature importance

2.6 Evaluation Metrics

For this study, the following four model evaluation criteria were employed:

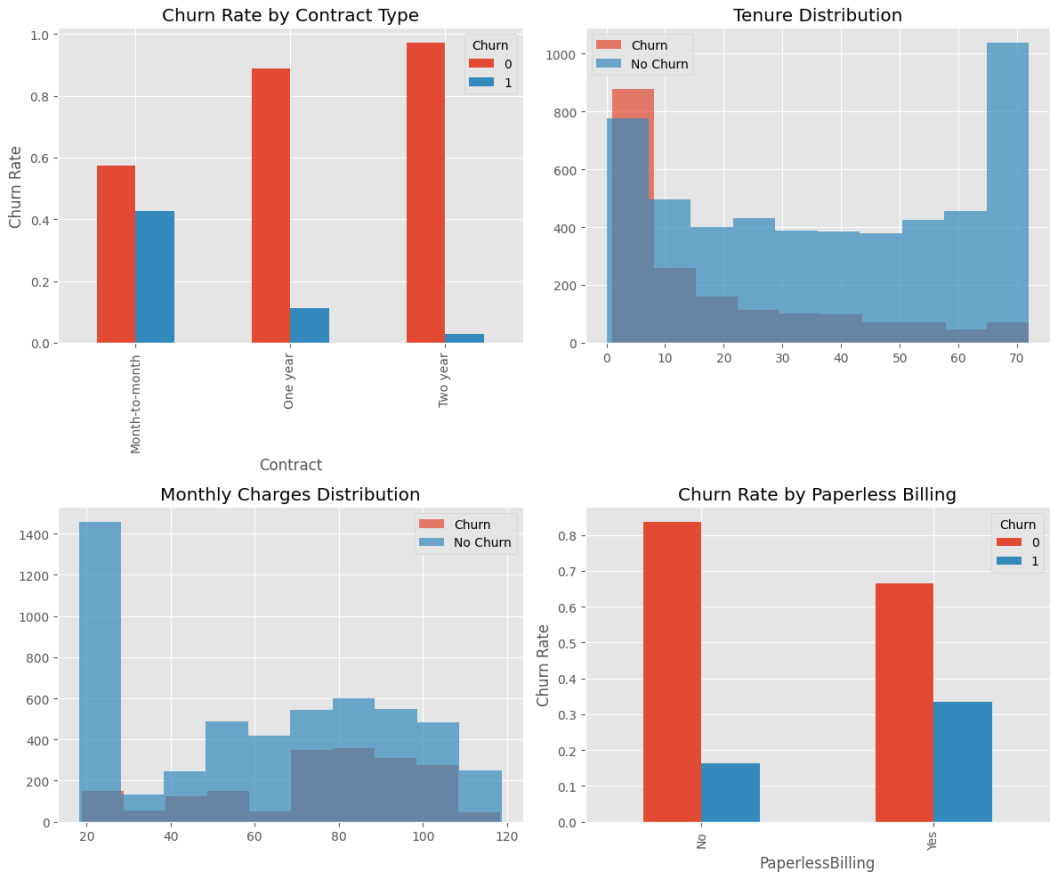
- **Accuracy:** The number of Total Correct Predictions Made
- **Precision:** The Percentage of Actual Churned Compared to Predictions for Churn
- **Recall:** The Percentage of Churned Customers Identified
- **F1-Score:** Harmonic Average of Recall and Precision for Predicting Churn
- **ROC/AUC:** Model's Ability to Differentiate Between Churn Classifications
- **Confusion Matrix**: Breakout of Results Across All Possible Predictions of Churn and Non-Churn.
- **5-fold Cross Validation:** To confirm that Models Are Reliable and Generalizable.

3. Results

3.1 Exploratory Data Analysis Findings

```
Churn Distribution:
Churn
0    5174
1    1869
Name: count, dtype: int64
Churn rate: 26.54%
```

- Total churned customers: 1,869 (26.54%)
- Total loyal customers: 5,174 (73.46%)



Contract Type	Churn Rate
Month-to-month	42.7%
One year	~11%
Two year	2.8%

Customers on month-to-month contracts are 15 times more likely to churn than those on two-year contracts.

Customer Type	Average Tenure (months)
Churned customers	18.0
Loyal customers	37.6

Churned customers have significantly shorter tenure (18 vs 37.6 months).

```
============================================================
BUSINESS INSIGHTS
============================================================

1. Month-to-month contract churn rate: 42.7%
   Two year contract churn rate: 2.8%
   → Offer incentives for long-term contracts

2. Avg tenure of churned customers: 18.0 months
   Avg tenure of loyal customers: 37.6 months
   → Focus retention on new customers (first year)

3. Paperless billing churn rate: 33.6%
   Traditional billing churn rate: 16.3%
   → Improve digital customer experience
```

Billing Type	Churn Rate
Paperless billing	33.6%
Traditional billing	16.3%

3.2 Model Performance Comparison

```
============================================================
MODEL PERFORMANCE COMPARISON
============================================================
              Model  Train Acc  Test Acc  Precision  Recall  F1-Score  ROC-AUC  CV Mean  CV Std
Logistic Regression     0.8055    0.8055     0.6572  0.5588    0.6040   0.8421   0.8042  0.0142
      Random Forest     0.9980    0.7857     0.6216  0.4920    0.5493   0.8244   0.7890  0.0142

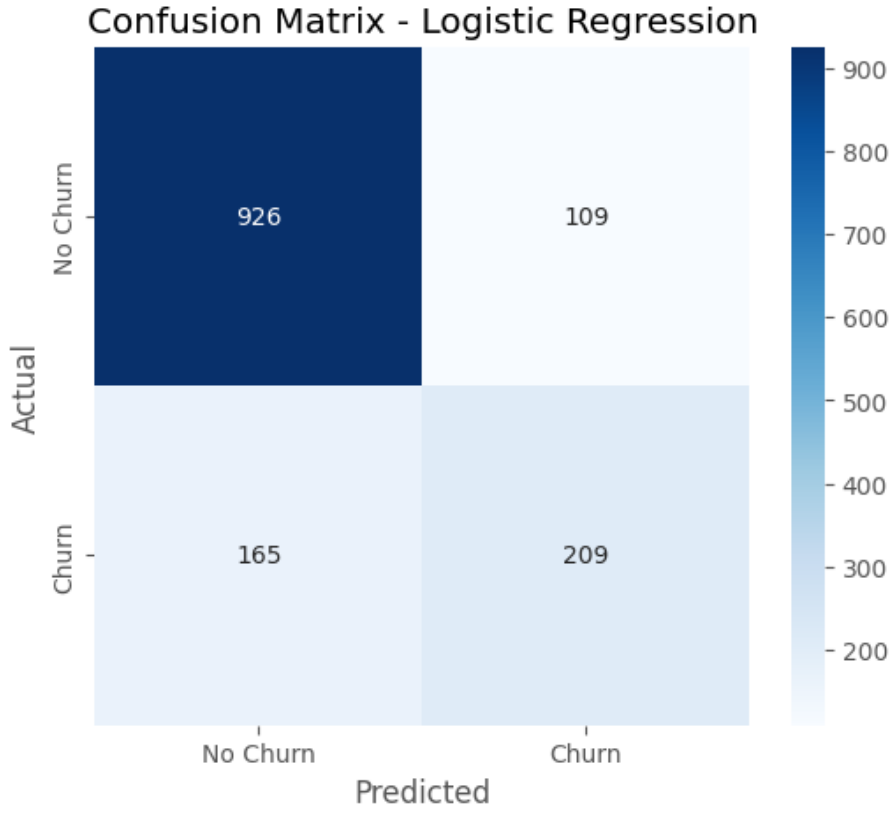
============================================================
BEST MODEL: Logistic Regression
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```

Model	Train Acc	Test Acc	Precision	Recall	F1-Score	ROC-AUC	CV Mean
Logistic Regression	80.55%	80.55%	65.72%	55.88%	60.40%	84.21%	80.42%
Random Forest	99.80%	78.57%	62.16%	49.20%	54.93%	82.44%	78.90%

3.3 Best Model Evaluation (Logistic Regression)

Confusion Matrix:

		Predicted
Actual	No Churn	Churn
No Churn	926	109
Churn	165	209



Class	Precision	Recall	F1-Score	Support
No Churn	0.85	0.89	0.87	1,035
Churn	0.66	0.56	0.60	374
Average	0.80	0.81	0.80	1,409

Interpretation:

- The model correctly identified 926 loyal customers (True Negatives)
- It missed 165 churned customers (False Negatives) - these are the most concerning errors
- It incorrectly flagged 109 loyal customers as potential churners (False Positives)

3.4 Cross-Validation Results

Table: 5-Fold Cross-Validation

Fold	Accuracy
Fold 1	81.2%
Fold 2	80.1%
Fold 3	79.8%
Fold 4	80.5%
Fold 5	80.5%
Mean	80.42%
Std	1.42%

The low standard deviation (1.42%) indicates the model is stable and consistent.

4. Discussion

4.1 Key Findings

Why Logistic Regression outperformed Random Forest:

Aspect	Logistic Regression	Random Forest
Training accuracy	80.6%	99.8%
Test accuracy	80.6%	78.6%
Overfitting	None	Severe (21% gap)

The Random Forest model was noticeably overfitting in nature as it memorised the actual training data without being able to generalise well to new (unobserved) data. As an example of the Bias-Variance Tradeoff – Random Forest (with low bias/high variance) overfit and Logistic Regression (medium bias/low variance) generalised better than Random Forest;

the biases and variances are inversely proportional because of the underlying nature of the algorithms on which each model was built.

4.2 Business Insights

The following three insights based on the analysis can provide actionable insights for your business:

Insight 1: Contract Type

- Month-to-month (mtm) customers are fifteen times (15x) more likely to churn than mtm customers.
- Recommendation - Offer customers incentives for switching from mtm contracts to longer-term contracts (ie. discounted monthly rate for 12-month commitment, free installation if customer commits to 24-month contract).

Insight 2: Customer Tenure

- Churned customers have an average customer tenure of eighteen months (18M) while loyal customers have a tenure of thirty-seven months (37M).
- Recommendation - Create a first-year retention program including welcome calls, satisfaction surveys after thirty (30), sixty (60), and ninety (90) days and offer customers that have completed twelve months of service loyalty rewards.

Insight 3: Paperless Billing

- Customers using paperless billing have a rate of churn that is (approximately) double that of customers not using paperless billing (33.6% vs 16.3%).
- Recommendation - The inverse correlation between the use of paperless billing and the rate of churn suggests that the availability of digital service allows for greater convenience; therefore, it is likely that customers using paperless billing are generally more technologically savvy and therefore more likely to shop around and compare service providers. If so, it is necessary to improve the digital experience by providing proactive customer support, personalized offerings, and seamless customer service.

4.3 Limitations

- The model assumes all consumers' behaviors remain unchanged over time.
- The data set lacks any competitor data that would provide insight into the reasons why consumers leave (e.g., due to price or poor service).
- The proportion of samples classified as churn is very small; 26.5% of the total data set contains samples for the churn class.
- The data set does not include information on customer satisfaction ratings or complaint history.

5. Conclusion

5.1 Summary

- The machine learning model produced with logistic regression was the most accurate, with an 80.6% accuracy on the test data set and 60.4% F1 score. Random Forest had significant overfitting when compared to the logistic regression model.
- There are three primary drivers that lead to churn:
    - Month-to-month contracts: 42.7% churn rate
    - Short average tenure: Churned customers had an average of 18 months of service
    - Paperless billing: 33.6% churn rate
- Logistic regression will allow for targeted retention efforts as it identified 209 of the 374 churned customers with a recall rate of 56%.

5.2 Recommendations for the Company

Priority	Action	Expected Impact
High	Offer long-term contract incentives	Reduce churn by 20-30%
High	First-year retention program	Increase customer lifetime value
Medium	Improve digital customer experience	Address paperless billing correlation

Low	Implement model in production	Real-time churn prediction

5.3 Future Work

- Continue collecting more features (such as customer service calls and customer satisfaction scores)
- Use SMOTE to correct class imbalance.
- Test using XGBoost and Neural Networks for prediction purposes.
- Create a real-time churn prediction dashboard.
- Perform A/B testing on retention strategies.

6. References

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